

California Polytechnic University, Pomona
B.S. in Chemical Engineering, 2028
GPA: 3.2/4.0

Jack Chobanian
Pomona, CA
831-710-7008
jachobanian@gmail.com
linkedin.com/in/jack-chobanian

Summary

Chemical Engineering student with hands-on production experience, process simulation, equipment troubleshooting, data analysis, and process design. Interested in refinery operations, manufacturing, and process engineering.

Work Experience

Student Tutor, Cal Poly Pomona

Sep. 2026 - Present

- Tutor first- and second-year students in calculus, linear algebra, differential equations, physics, chemistry, and materials science.
- Explain challenging STEM concepts and adapt lessons to different learning styles
- Help students develop effective study habits, build confidence, and strengthen problem-solving skills.

Production Engineering Intern, Rustique Wines

Jun. 2024 - Dec. 2024

- Monitored and documented temperature, pH, and fermentation conditions for 10+ tanks, using Microsoft Excel to organize production data, track trends, and identify process deviations.
- Operated and troubleshoot pumps, tanks, valves, and filtration equipment, helping maintain production schedules and minimize equipment downtime.
- Followed strict safety, sanitation, and quality control procedures while supporting the production of approximately 10,000 gallons of finished product.

Sales Associate, Trader Joe's

Aug. 2024 - Jan. 2026

- Managed beverage inventory and submitted stock orders based on product demand and availability.
- Collaborated with team members to stock products, maintain cleanliness, and meet daily operational goals.

Relevant Projects

Biodiesel Production and Operations | Personal Project

Jan. 2025

- Prepared feedstock and reactants for biodiesel production
 - Used MATLAB to calculate the required quantities of reactants for processing 3 L of used cooking oil.
 - Measured 600 mL of methanol and 10.5 g of sodium hydroxide catalyst to create the methoxide solution.
- Conducted transesterification reaction
 - Converted triglycerides in the oil into biodiesel while producing glycerol as a byproduct.
- Separated and refined the finished product
 - Allowed the mixture to settle before separating the heavier glycerol phase from the biodiesel layer.
 - Applied material balances, stoichiometry, phase separation, and process-operation principles relevant to petroleum refining and fuel production.

Shell and Tube Heat Exchanger Design | Personal Project | Onshape

Jul. 2026

- Designed a shell-and-tube heat exchanger in Onshape using mechanical design and heat-transfer principles.
 - Defined the shell, tube bundle, flow arrangement, and major assembly dimensions.
- Created a detailed CAD assembly with labeled components and engineering dimensions for technical presentation.
 - Modeled a 120-inch shell, 116-inch tubes, 69-tube bundle, tube sheets, channel heads, flanges, and nozzles.
- Used basic mechanical design concepts to model a functional and realistic heat exchanger assembly.

Technical Skills

MATLAB | Microsoft Excel | PRO/II | Python | Onshape CAD